The Behavioural Evaluation Framework for AI Judgement Systems

Imogen Hull

Independent Researcher – AI Behaviour & Reliability

Beyond the Average Research Series

Version 1.0 – March 2026

Working Paper
Beyond the Average Research Series
March 2026

Executive Summary

Large language models are increasingly used to evaluate text, classify information and assist in decisions that may affect individuals or organisations. Many current evaluation approaches assess the quality of individual outputs, examining whether a response appears accurate, unbiased or well explained.

However, a single output reveals very little about how an AI system actually behaves. Language-based systems generate responses probabilistically, and their reasoning may vary when the same task is repeated. As a result, systems that appear reliable when observed once may exhibit unstable judgement patterns across repeated executions.

In many fields that deal with complex systems, reliability is not assessed through a single observation. Engineers, auditors and safety specialists examine behaviour across repeated operations, monitoring signals and comparing outcomes across independent systems to understand how systems behave over time.

This paper introduces the Behavioural Evaluation Framework, an approach for examining how AI judgement systems behave when observed repeatedly under controlled conditions. The framework proposes three complementary perspectives for evaluation: repeated execution of the same task, examination of internal signals such as confidence or agreement and comparison between independent systems performing the same evaluation.

Experiments examining how AI systems evaluate job advertisements for potential age bias illustrate how repeated observation reveals behavioural patterns that single outputs conceal. Some judgements remain stable across runs, while others converge, drift or fragment.

The Behavioural Evaluation Framework therefore reframes AI evaluation as the study of behavioural stability rather than the inspection of isolated outputs.

Abstract

Most evaluation methods for large language models focus on individual outputs, assessing whether a model’s response is accurate, unbiased or well explained. However, single outputs reveal little about how a system behaves. Language-based AI systems are probabilistic and their responses may vary across repeated executions of the same task.

This paper introduces the **Behavioural Evaluation Framework**, an approach for examining how AI judgement systems behave when observed repeatedly under controlled conditions. Drawing on principles from reliability engineering, safety engineering, software fault tolerance and financial auditing, the framework proposes three complementary perspectives for evaluating AI judgement behaviour: repeated execution of the same task, examination of internal signals such as confidence or agreement and comparison between independent systems performing the same evaluation.

Using experiments involving the evaluation of job advertisements for potential age bias, the paper illustrates how repeated observation reveals behavioural patterns that single outputs conceal. The Behavioural Evaluation Framework reframes AI evaluation as the observation of behavioural stability rather than the inspection of isolated answers.

Keywords

AI evaluation
model reliability
behavioural evaluation
judgement stability
AI auditing
system behaviour

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1. Introduction

Large language models increasingly perform evaluative tasks. They classify information, interpret text, identify risks and assist in decisions that may influence outcomes for individuals or organisations. Evaluations of such systems often focus on the quality of individual outputs. A model may be judged by whether its response appears correct, unbiased or well explained.

Yet this approach provides limited insight into system behaviour. Language-based AI systems generate responses probabilistically and their reasoning may vary across repeated executions of the same task. A single output therefore reveals little about how stable the system's judgement actually is.

In most fields that deal with complex systems, reliability is not assessed through isolated observations. Engineers, auditors and safety researchers evaluate behaviour across repeated operations, examining patterns that emerge over time. Variation across repeated executions is treated as evidence about system behaviour rather than as an anomaly.

The Behavioural Evaluation Framework described in this paper emerged from a series of experimental studies examining how AI systems evaluate job advertisements for potential age bias. These experiments involved repeated execution of the same evaluative task across multiple language models in order to observe how judgements, explanations and internal signals behaved when examined repeatedly under controlled conditions.

This work forms part of a broader research programme examining behavioural auditing of AI judgement systems, developed within the **BOUNDARY approach** (Behavioural Oversight for Uncertainty, Noise, Drift and Reliability).

Rather than focusing solely on single outputs, the Behavioural Evaluation Framework proposes observing behavioural patterns that emerge across repeated tasks, internal monitoring signals and comparisons between independent systems.

The Behavioural Evaluation Framework defines AI system reliability as the stability of judgement behaviour observed across repeated execution of the same task, examination of internal system signals and comparison between independent systems performing the same evaluation.

This paper makes three contributions. First, it identifies the limitations of single-output evaluation when assessing AI judgement systems. Second, it synthesises concepts from reliability engineering, safety engineering and auditing to frame AI evaluation as a behavioural problem. Third, it introduces the Behavioural Evaluation Framework as a simple structure for observing judgement behaviour across repeated execution, internal signals and independent system comparison.

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2. Reliability and Behaviour in Complex Systems

In engineering disciplines, system reliability is typically assessed through repeated operation under controlled conditions. Components are tested multiple times to determine whether their behaviour remains stable or begins to vary.

When repeated operations produce different outcomes under identical conditions, that variation is treated as a signal. Engineers do not assume that one result is correct and the others are errors. Instead, variation itself becomes evidence that the system requires further examination.

Reliability engineering therefore emphasises observation across repeated executions rather than reliance on a single outcome (Ebeling, 2010).

Similar approaches appear in safety engineering, where monitoring signals and feedback loops are used to understand how systems behave during operation. Leveson’s STAMP framework similarly emphasises analysing system interactions rather than isolated failures (Leveson, 2011).

In software engineering, early fault-tolerant architectures explored the idea of independent implementations performing the same task in parallel. Differences between implementations could reveal faults or inconsistencies in system behaviour (Avizienis & Chen, 1977).

Financial auditing follows a comparable logic. Auditors rarely rely on a single transaction when evaluating the reliability of financial systems. Instead, they examine patterns across many transactions to determine whether behaviour remains consistent or begins to diverge (Arens, Elder & Beasley, 2017).

Across these disciplines, reliability emerges from **patterns of behaviour rather than isolated events**.

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3. The Single Output Problem in AI Evaluation

Many current approaches to AI evaluation remain centred on individual outputs. Benchmarks typically examine whether a model produces a correct answer, an unbiased classification or a well-structured explanation.

However, language models behave probabilistically. Their responses are sensitive to context, prompt structure and internal sampling processes. As a result, the same model may produce different responses when the same task is repeated.

When evaluation focuses on a single output, such behavioural variation remains invisible. A model may appear reliable when observed once while displaying unstable judgement patterns across repeated runs.

Recent work on AI accountability has emphasised transparency and documentation of model behaviour through mechanisms such as model cards and auditing frameworks (Mitchell et al., 2019; Raji et al., 2020). These approaches improve visibility into system design and intended use but do not necessarily reveal how judgement behaviour varies across repeated evaluations.

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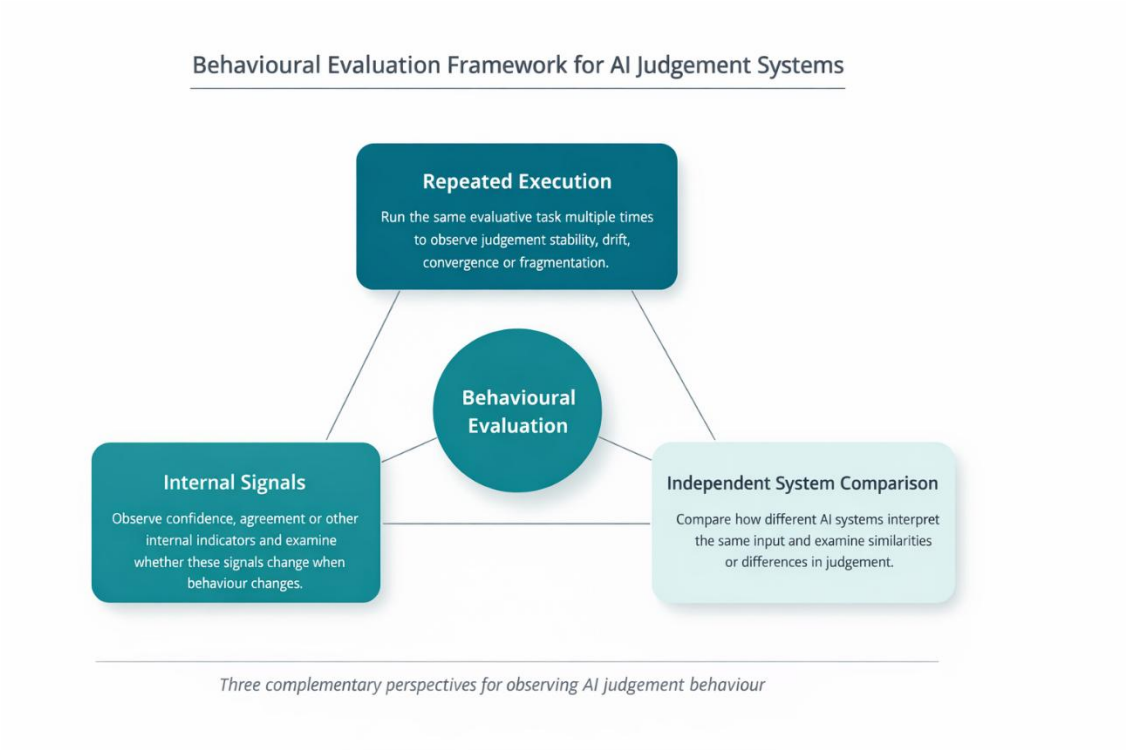
4. The Behavioural Evaluation Framework

The Behavioural Evaluation Framework proposes examining AI judgement systems from three complementary perspectives: repeated execution of the same task, examination of internal signals and comparison between independent systems performing the same evaluation.

Figure 1 illustrates the three complementary perspectives of the Behavioural Evaluation Framework.

Figure 1 – Behavioural Evaluation Framework for AI Judgement Systems



A framework for evaluating behavioural stability in AI judgement systems

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4.1 Repeated Execution

The same evaluative task is executed multiple times under controlled conditions. Observing repeated outputs allows analysts to examine whether judgements remain stable or begin to vary.

Across repeated runs, behavioural patterns may emerge including stable judgement, gradual convergence, drift in reasoning or fragmentation in which different interpretations appear across executions.

4.2 Internal Signals

Many AI systems produce internal indicators such as confidence scores, agreement measures or other evaluation signals. These signals are often interpreted as indicators of reliability.

However, their usefulness depends on whether they respond meaningfully when behaviour changes. Examining how these signals vary across repeated executions helps determine whether they reflect underlying system behaviour or remain static even when outputs vary.

4.3 Independent System Comparison

A third perspective involves comparing the behaviour of different AI systems performing the same evaluation task.

Independent systems may recognise similar signals within an input yet reach different conclusions. These differences may arise from variations in training data, model architecture or decision thresholds.

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5. Experimental Illustration

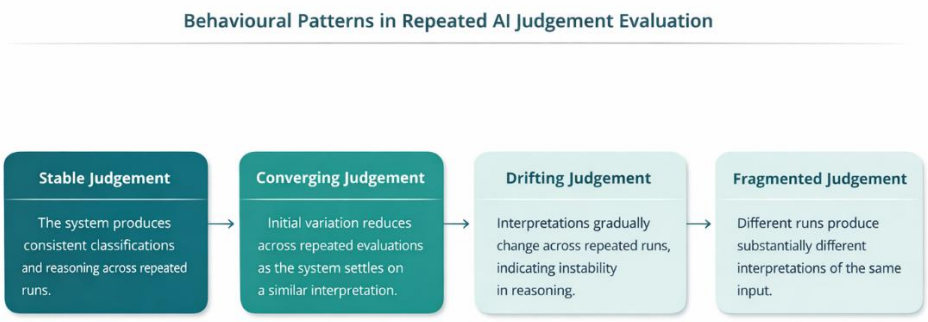
The framework emerged from experiments examining how AI systems evaluate job advertisements for potential age-coded language.

Repeated evaluation of the same advertisement was performed across multiple runs and across different large language models. These experiments were designed to observe how classification outcomes, explanations and internal signals behaved when the same evaluative task was repeated under controlled conditions across multiple runs and models.

Across repeated runs, several behavioural patterns became visible. Some judgements remained stable, producing similar classifications and reasoning across executions. In other cases, interpretations varied subtly, with explanations shifting in emphasis even when the final classification remained similar.

Figure 2 illustrates the types of behavioural patterns that may emerge when AI judgement tasks are repeated.

Figure 2 – Behavioural Patterns in Repeated AI Judgement Evaluation



Observed behavioural patterns when AI judgement tasks are repeated under controlled conditions

Comparisons between models performing the same task revealed another pattern. Systems often identified similar textual cues yet applied different thresholds when forming a judgement.

These observations illustrate how behavioural variation becomes visible only when evaluation is repeated.

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6. Behavioural Evaluation and AI Observability

A growing ecosystem of AI observability tools allows developers to monitor model behaviour during deployment. Platforms such as Comet ML, Weights & Biases and LangSmith record prompts, outputs and evaluation results so that teams can study how systems perform in operational environments.

These tools are designed primarily to monitor behaviour after systems have been deployed.

The Behavioural Evaluation Framework addresses an earlier stage of evaluation. Before monitoring dashboards are required, repeated observation can reveal how stable a system's judgement is when performing the same task under controlled conditions.

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7. Discussion

The Behavioural Evaluation Framework reframes AI evaluation as the observation of behavioural patterns rather than the inspection of isolated outputs.

By examining repeated executions, internal signals and comparisons across independent systems, evaluators can gain insight into how AI judgement forms and whether it remains stable across repeated observations.

This perspective aligns AI evaluation with established practices used in reliability engineering, safety analysis and auditing.

The framework introduced here is conceptual rather than prescriptive, and further work will be required to develop systematic testing protocols and quantitative measures for behavioural evaluation.

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8. Conclusion

Evaluating AI systems through behavioural observation places the study of AI judgement within a broader tradition of examining complex systems through repeated observation, monitoring signals and independent comparison.

Rather than relying solely on isolated outputs, evaluators can observe how judgement behaves when tasks are repeated, signals are examined and independent systems perform the same evaluation.

In systems that influence decisions about people, understanding how judgement behaves across repeated observations may be as important as the answer produced in any single instance.

Evaluating AI systems through behavioural observation places the study of AI judgement within a broader tradition of examining complex systems through repeated observation, monitoring signals and independent comparison. Rather than relying solely on isolated outputs, evaluators can observe how judgement behaves when tasks are repeated, signals are examined and independent systems perform the same evaluation. In systems that influence decisions about people, understanding the behavioural stability of AI judgement may be as important as evaluating the correctness of any individual output.

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Citation

Hull, I. (2026). *The Behavioural Evaluation Framework for AI Judgement Systems.* Beyond the Average Research Series. Zenodo.